TIC 394942247

Eclipsing binary candidate dossier - Star Vetter

CLEAN

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BLS EPHEMERIS
  Period        2.432054 d
  Epoch (t0)    1684.0694 BTJD
  Duration      0.1200 d
  Depth         29.268 %  (292,675 ppm)
  SNR           908.9

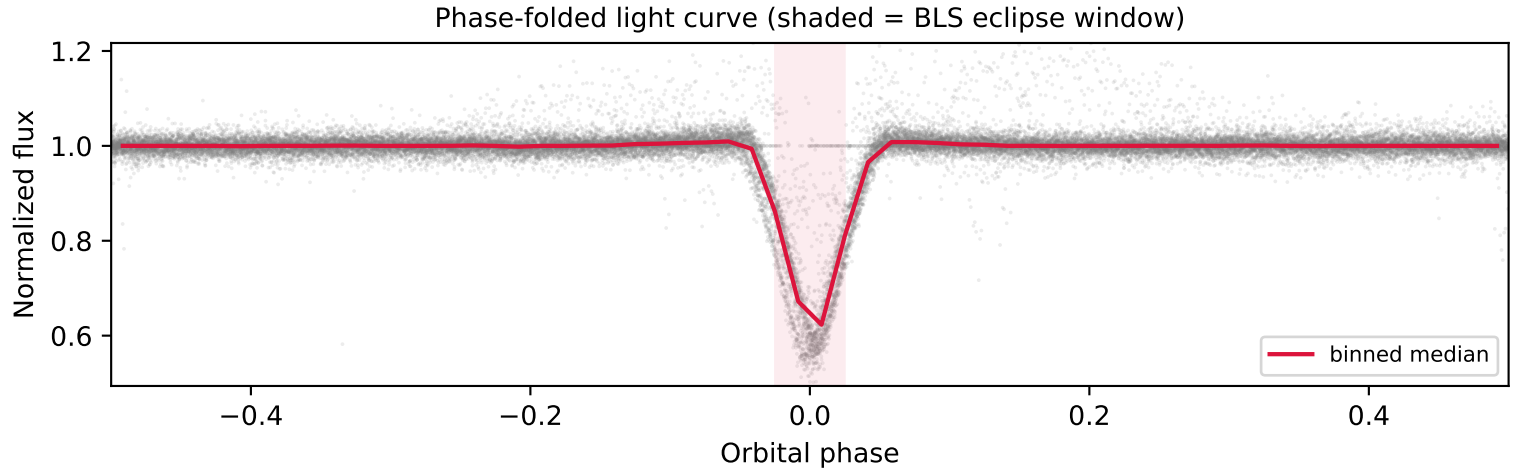
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      64  (gate >= 3)
  Robust depth ratio     1.04  (gate >= 0.50)
  Robust depth sigma     698.8  (gate >= 7.0)
  Odd/even z-score       0.73  (consistent)
  Secondary eclipse      not detected
```

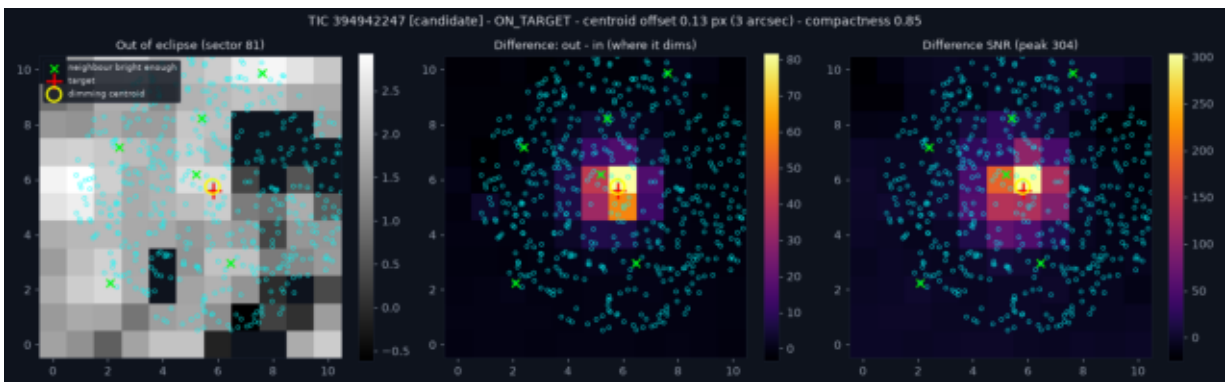
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 81    light-curve dip in this sector: 427.8 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   303.87   (gate >= 5.0)      Compactness   0.85   (gate >= 0.50)
  Dimming centroid offset     0.13 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 394942233
    0.76 px away, dTmag +0.06 - farther from the dimming than the target
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No caveats - clean detection on every check.